N-IOC TO RULE THEM ALL

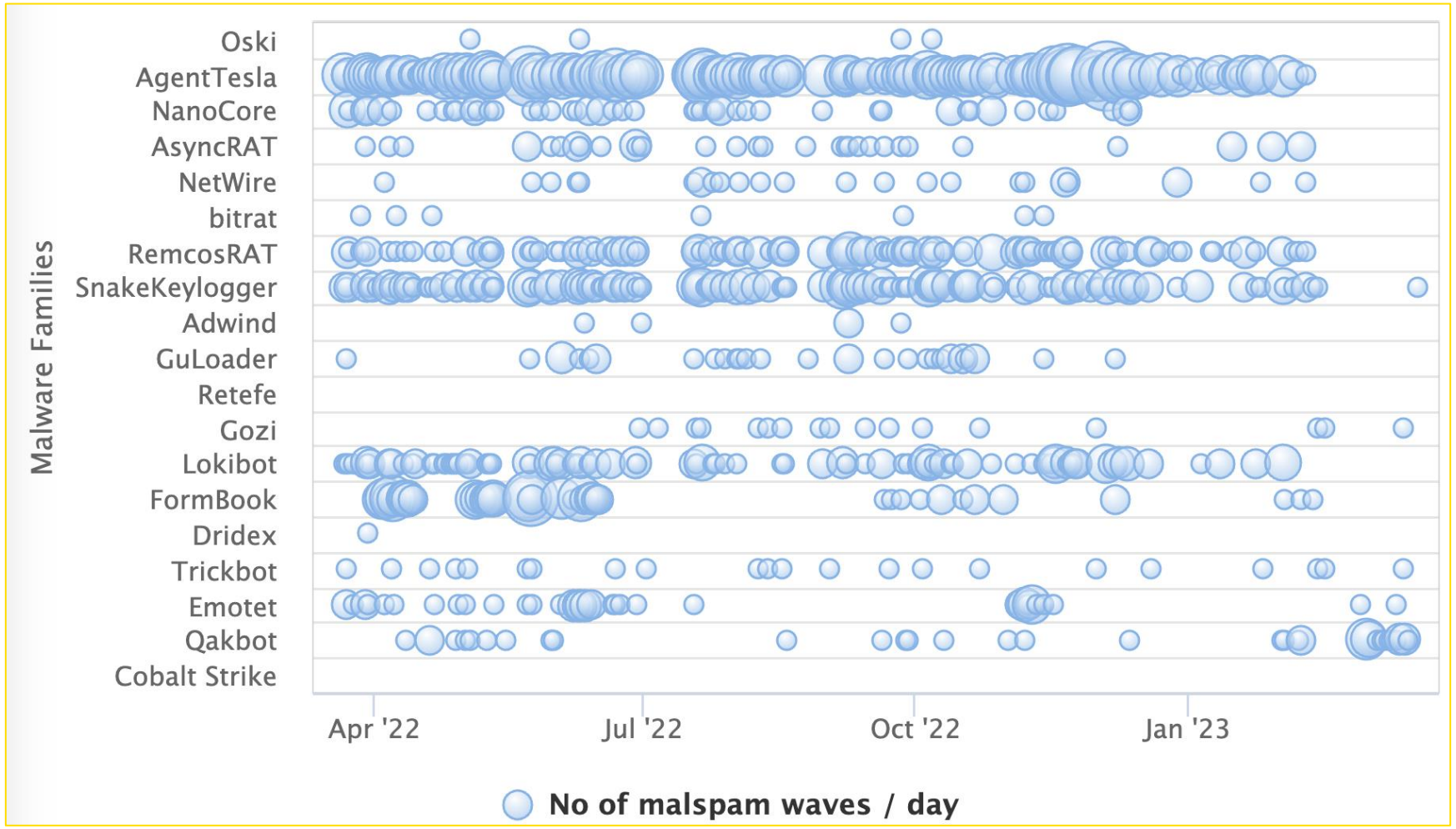
InfoGuard
SWISS CYBER SECURITY

ABOUT ME (STEPHAN BERGER)

Head of Investigations at InfoGuard AG

- Spending probably too much time on
- Luckily I have an excellent team covering my back
- Master of musical song references
- SysAdmin / SOC-Analyst / CSIRT / Consultant

MALWARE EVENTS IN SWITZERLAND



Source: govcert.ch

TOP 10 MALWARE SAMPLES (*GOVCERT.CH*)

TOP 10 MALWARE SAMPLES
analyzed by GovCERT.ch in April 2022

		3 weeks ago	last week
1	Agent Tesla aka AgenTesla, AgentTesla, Negasteal	41.5%	45.5%
2	Xloader / Formbook aka Formbook	30.9%	25.8%
3	404 Keylogger aka 404KeyLogger, Snake Keylogger	10.6%	9.20%
4	Loki Password Stealer (PWS) aka Burkina, Loki, LokiBot	1.32%	5.06%
5	Nanocore RAT aka Nancrat, NanoCore	3.08%	2.76%
6	Emotet aka Geodo, Heodo	1.65%	5.06%
7	Ave Maria aka AVE_MARIA, AveMariaRAT, Warzone RAT		NEW 1.84 %
8	CloudEyE aka GuLoader, vbdropper	0.88%	0.92%
·	AsyncRAT	0.44%	0.92%
10	QakBot aka Pinkslipbot, Qbot, Quakbot	1.20%	1.46 %

Names and aliases according to Malpedia. The percentages show the proportion of analyses from the total number of analyses.

NCSC / GovCERT.ch CC BY-SA 2.0

TOP 10 MALWARE SAMPLES
analyzed by GovCERT.ch in May 2022

		3 weeks ago	last week
1	Agent Tesla aka AgenTesla, AgentTesla, Negasteal	47.7%	39.5%
2	Xloader / Formbook aka Formbook	17.8%	23.0%
3	404 Keylogger aka 404KeyLogger, Snake Keylogger	12.5%	12.7 %
4	Loki Password Stealer (PWS) aka Burkina, Loki, LokiBot	7.68%	8.46%
5	Ave Maria aka AVE_MARIA, AveMariaRAT, Warzone RAT		NEW 5.17 %
6	Remcos aka RemcosRAT, Remvio, Socmer	3.36%	3.76%
7	Emotet aka Geodo, Heodo	0.96%	3.35%
8	QakBot aka Pinkslipbot, Qbot, Quakbot	1.23%	2.29%
9	AsyncRAT		NEW 0.94%
10	RedLine Stealer	0.96%	0.47%

Names and aliases according to Malpedia. The percentages show the proportion of analyses from the total number of analyses.

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TOP 10 MALWARE SAMPLES
analyzed by GovCERT.ch in June 2022

		3 weeks ago	last week
1	Agent Tesla aka AgenTesla, AgentTesla, Negasteal	37.1%	33.0%
2	Xloader / Formbook aka Formbook	20.2%	21.6 %
3	404 Keylogger aka 404KeyLogger, Snake Keylogger	6.90%	18.0 %
4	Remcos aka RemcosRAT, Remvio, Socmer	2.30%	6.00%
·	Emotet aka Geodo, Heodo	7.36%	6.00%
6	Ave Maria aka AVE_MARIA, AveMariaRAT, Warzone RAT		NEW 3.00%
·	Loki Password Stealer (PWS) aka Burkina, Loki, LokiBot	12.4%	3.00%
8	RedLine Stealer	1.38%	2.40%
·	QakBot aka Pinkslipbot, Qbot, Quakbot	1.54%	2.40%
10	Nanocore RAT aka Nancrat, NanoCore	0.92%	1.20 %

Names and aliases according to Malpedia. The percentages show the proportion of analyses from the total number of analyses.

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Source: govcert.ch

TOP 10 MALWARE FAMILIES (*GOVCERT.CH*)

- Agent Tesla
- Formbook
- 404 Keylogger
- Loki Password Stealer
- Ave Maria RAT
- Remcos
- Nanocore RAT
- QakBot
- Redline Stealer
- Netwire RAT

Source: April 2022 – December 2022

REVERSE ENGINEERING..

WORK SMARTER, NOT HARDER

SAMPLES: VX-UNDERGROUND

Bazaar.2022.09.7z	6596122460	2022-10-17 22:49:31
Bazaar.2022.10.7z	10390770357	2022-11-01 12:02:31
Bazaar.2022.11.7z	10385352800	2022-12-02 06:57:17
Bazaar.2022.12.7z	10117808288	2023-01-01 07:02:50
Bazaar.2023.01.7z	10584395712	2023-02-01 08:36:56
Bazaar.2023.02.7z	9919966899	2023-03-01 23:14:34
Bazaar.2023.03.7z	6219214521	2023-04-05 22:06:25

IOC #1: DEFENDER EXCLUSIONS

```
Add-MpPreference -ExclusionPath
"C:\Users\user\AppData\Roaming\qLCTUoBHeGs.exe"
# Nanocore RAT
```

AT	FB	404	LPS	AM	RC	NCR	QB	RLS	NWR
✅	✅	✅	❌	✅	❌	✅	✅	✅	❌

ARTIFACTS: EVENT 5007

Event 5007, Windows Defender

General | Details

Microsoft Defender Antivirus Configuration has changed. If this is an unexpected event you should review the settings as this may be the result of malware.

Old value:

New value: HKLM\SOFTWARE\Microsoft\Windows Defender\Exclusions\Paths\C:\Users\user\AppData\Roaming\qLCTUoBHeGs.exe = 0x0

Log Name:	Microsoft-Windows-Windows Defender/Operational		
Source:	Windows Defender	Logged:	3/23/2023 8:08:56 PM
Event ID:	5007	Task Category:	None
Level:	Information	Keywords:	

ARTIFACTS: REGISTRY

Administrator: Command Prompt

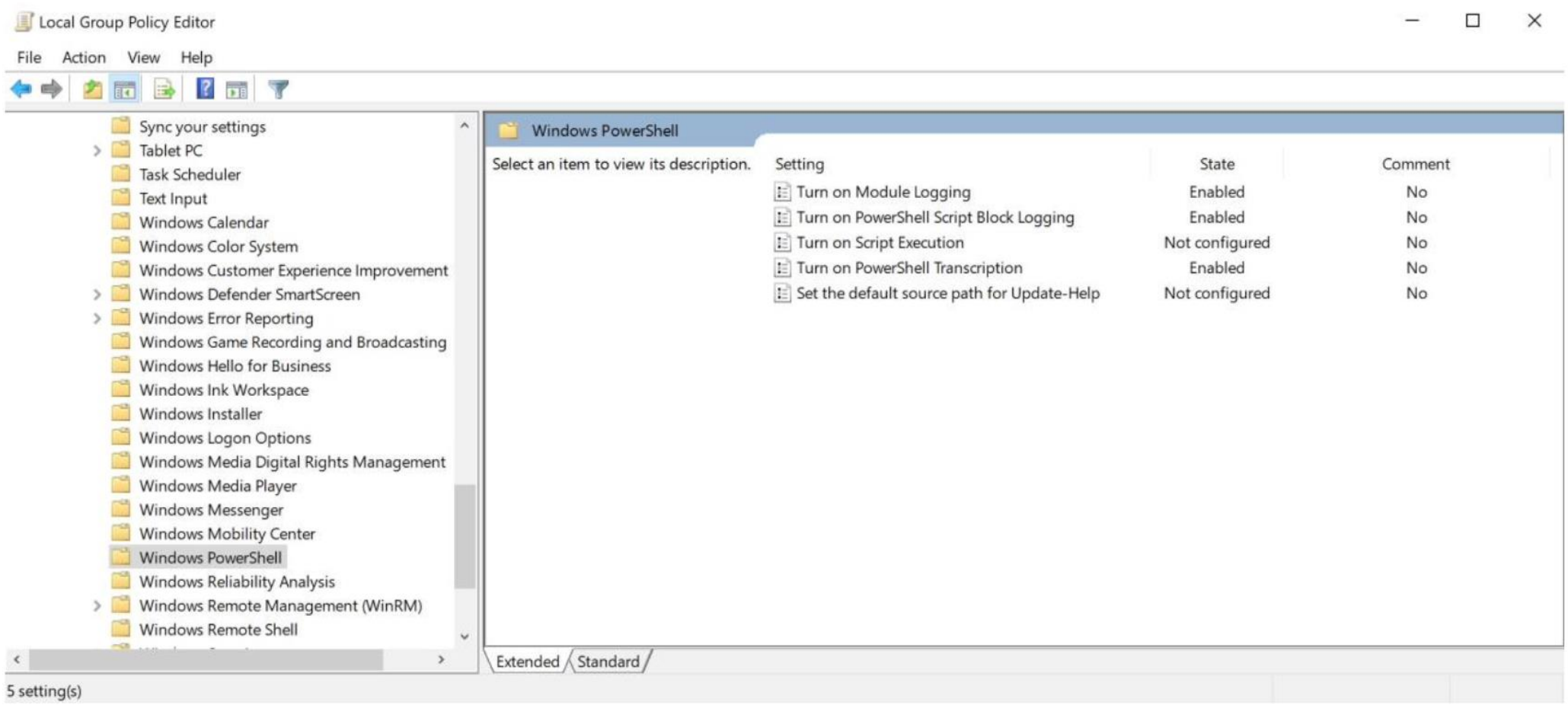
```
C:\>reg query "HKLM\SOFTWARE\Microsoft\Windows Defender\Exclusions\Paths"

HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows Defender\Exclusions\Paths
    C:\Users\user\AppData\Roaming\qLCTUoBHeGs.exe    REG_DWORD    0x0
```

IOC #2: POWERSHELL

```
New-ItemProperty -Path
'HKCU:\SOFTWARE\Microsoft\Windows\CurrentVersion\Run'
-Name 'RZTHXHelper' -Value
'"C:\Users\user\AppData\Roaming\Microsoft\Windows\
Recent\RZTHX\RZTHXHelper.exe"' -PropertyType 'String'
# Ave Maria RAT
```

AT	FB	404	LPS	AM	RC	NCR	QB	RLS	NWR
(✅)	❌	✅	❌	(✅)	(✅)	(✅)	❌	✅	✅

ARTIFACTS: EVENT 4104



ARTIFACTS: EVENT 4104

Event 4104, PowerShell (Microsoft-Windows-PowerShell)

General | Details

Creating Scriptblock text (1 of 1):

```
New-ItemProperty -Path 'HKCU:\SOFTWARE\Microsoft\Windows\CurrentVersion\Run' -Name 'RZTHXHelper' -Value '"C:
\Users\user\AppData\Roaming\Microsoft\Windows\Recent\RZTHX\RZTHXHelper.exe"' -PropertyType 'String'
```

ScriptBlock ID: 95b67aad-91f9-4b1f-b211-0b251060d805
Path:

ARTIFACTS: EVENT 4104

Nasreddine Bencherchali
@nas_bench

PowerShell has a list of suspicious keywords. If found in a script block an automatic 4104 event will be generated regardless of logging policy :) (True for both PWSH 5/7)

Look for EID 4104 with Level 3 (Warning)

Full List: gist.github.com/nasbench/50cd0...

```
uick check for script blocks that may have suspicious content. If this
s true, we log them to the event log despite event log settings.
rnal static string CheckSuspiciousContent(Ast scriptBlockAst)

var foundSignature = SuspiciousContentChecker.Match(scriptBlockAst.Extent.Text);
if (foundSignature != null)
{
    return foundSignature;
}

if (scriptBlockAst.HasSuspiciousContent)
```

```
{
    // Calling Add-Type
    case 3012981990: return "Add-Type";
    case 3359423881: return "DllImport";

    // Doing dynamic assembly building / method indirection
    case 2713126922: return "DefineDynamicAssembly";
    case 2407049616: return "DefineDynamicModule";
    case 3276870517: return "DefineType";
    case 419507039: return "DefineConstructor";
    case 1370182198: return "CreateType";
```

IOC #3: SCHEDULED TASKS

```
schtasks.exe /Create /RU NT AUTHORITY\SYSTEM /tn
ouqjustt /tr regsvr32.exe -s
"C:\Users\User\Desktop\FIRST\qbot.dll" /SC ONCE /Z
/ST 16:40 /ET 16:52
#QakBot
```

AT	FB	404	LPS	AM	RC	NCR	QB	RLS	NWR
✅	✅	✅	❌	✅	✅	✅	✅	✅	✅

ARTIFACTS: C:/WINDOWS/SYSTEM32/TASKS/*

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Artifact details

Windows.System.TaskScheduler

Type: client

The Windows task scheduler is a common mechanism that malware uses for persistence. It can be used to run arbitrary programs at a later time. Commonly malware installs a scheduled task to run itself periodically to achieve persistence.

This artifact enumerates all the task jobs (which are XML files). The artifact uploads the original XML files and then analyses them to provide an overview of the commands executed and the user under which they will be run.

Parameters

Name	Type	Default	Description
TasksPath		c:/Windows/System32/Tasks/**	

ARTIFACTS: TASKS WITH COMMAND & LAUNCHSTRING

✔	F.CGTB84UHE4H50	Windows.System.TaskScheduler	2023-04-15T14:26:27Z	2023-04-15T14:27:16Z	stbe

C:\Windows\System32\Tasks\OneDrive Standalone Update Task-S-1-5-21-454062999-803709901-3569341773-1001	%localappdata%\Microsoft\OneDrive\OneDriveStandaloneUpdater.exe	
C:\Windows\System32\Tasks\npcapwatchdog	C:\Program Files\Npcap\CheckStatus.bat	
C:\Windows\System32\Tasks\rvocgveq	regsvr32.exe	-s "C:\Users\User\Desktop\FIRST\qbot.dll"
C:\Windows\System32\Tasks\Microsoft\VisualStudio\VSIX Auto Update	C:\Program Files (x86)\Microsoft Visual Studio\Installer\resources\app\ServiceHub\Services\Microsoft.VisualStudio.Setup.Service\VSIXAutoUpdate.exe	

ARTIFACTS: EVENT LOGS

SECURITY.EVTX

- 4698: A scheduled task was created
- 4699: A scheduled task was deleted
- 4700: A scheduled task was enabled
- 4701: A scheduled task was disabled
- 4702: A scheduled task was updated

MICROSOFT-WINDOWS-TASKSCHEDULER%4OPERATIONAL.EVTX

- 140: User <user> updated Task Scheduler task <task_name>
- 141: User <user> deleted Task Scheduler task <task_name>

IOC #4: RUN KEYS

```
HKCU\Software\Microsoft\Windows\CurrentVersion\Run

Fntssqo

"C:\Users\user\AppData\Roaming\Hxkotn\Fntssqo.exe"

# 404 Keylogger
```

AT	FB	404	LPS	AM	RC	NCR	QB	RLS	NWR
☒	✓	✓	☒	✓	✓	✓	✓	✓	✓

ARTIFACTS: SYSMON EVENT ID 13 - REGISTRY SET

Event 13, Sysmon

General | Details

```
Registry value set:
RuleName: T1060,RunKey
EventType: SetValue
UtcTime: 2023-04-16 10:17:41.925
ProcessGuid: {f0ddfc6e-cb45-643b-5113-000000001d00}
ProcessId: 844
Image: C:\Users\malmoeb\AppData\Local\Temp\cckgcf.exe
TargetObject: HKU\S-1-5-21-4008431687-740994179-4145981840-1000\SOFTWARE\Microsoft\Windows\CurrentVersion\Run\hhtktvn
Details: C:\Users\malmoeb\AppData\Roaming\gswccl\ratotpvvsmo.exe
```

ARTIFACTS: WHEN IN DOUBT..

```
KeyGlobs
HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows\CurrentVersion\Run*
HKEY_LOCAL_MACHINE\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\Run*
HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows\CurrentVersion\Policies\Explorer\Run*
HKEY_USERS*\SOFTWARE\Microsoft\Windows\CurrentVersion\Run*
HKEY_USERS*\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\Run*
HKEY_USERS*\SOFTWARE\Microsoft\Windows\CurrentVersion\Policies\Explorer\Run*
```

```
/*
# Windows.Sys.StartupItems
*/
SELECT * FROM source(artifact="Windows.Sys.StartupItems") where Details =~ "AppData"
```

Windows.Sys.StartupItems

Name	OSPath	Details
Microsoft Edge Update	HKEY_USERS\S-1-5-21-4008431687-740994179-4145981840-1000\SOFTWARE\Microsoft\Windows\CurrentVersion\Run\Microsoft Edge Update	"C:\Users\malmoeb\AppData\Local\Microsoft\EdgeUpdate\1.3. EdgeUpdateCore.exe"
OneDrive	HKEY_USERS\S-1-5-21-4008431687-740994179-4145981840-1000\SOFTWARE\Microsoft\Windows\CurrentVersion\Run\OneDrive	"C:\Users\malmoeb\AppData\Local\Microsoft\OneDrive\OneDri /background
hhtktvn	HKEY_USERS\S-1-5-21-4008431687-740994179-4145981840-1000\SOFTWARE\Microsoft\Windows\CurrentVersion\Run\hhtktvn	C:\Users\malmoeb\AppData\Roaming\gswccl\ratotpvvsmo.exe

IOC #5: STARTUP FOLDER

```
C:\Users\user\AppData\Roaming\Microsoft\Windows\Start
Menu\Programs\Startup\h2C08rJjFc.exe
# 404 Keylogger
```

AT	FB	404	LPS	AM	RC	NCR	QB	RLS	NWR
☒	✅	✅	☒	☒	☒	☒	☒	☒	✅

ARTIFACTS: AUTORUNS

Artifact details

Windows.Sysinternals.Autoruns

Type: client

Uses Sysinternals autoruns to scan the host.

Note this requires syncing the sysinternals binary from the host.

Tools

- Autorun_x86
- Autorun_amd64

ARTIFACTS: AUTORUNS

Artifact Collection | Uploaded Files | Requests | Results | Log | Notebook

```
/*
# Windows.Sysinternals.Autoruns
*/
SELECT * FROM source(artifact="Windows.Sysinternals.Autoruns")
where `Entry Location` =~ "Startup"
```

1922081009385 1	HKCU\Software\Microsoft\Windows\CurrentVersion\Explorer\Shell Folders\Startup	h2C08rJjFc.exe	enabled	Logon	DESKTOP-6C00RSU\malmoeb	Ertogo	Ertogo	c:\users\malmoeb\appdata\roaming\microsoft\windows\start menu\programs\startup\h2c08rjjfc.exe	0.0.0.0	C:\Users\malmoeb\AppData\Roaming\Microsoft\Windows\Start Menu\Programs\Startup\h2C08rJjFc.exe	312869 2094AB D246C8 6A0B91 B48741 8E	6301515 9C389ED 86D8B11 C58A0B5 6382404 71EA4

IOC #6: APPDATA

```
File Writte:
"C:\Users\user\AppData\Local\Temp\mnsywlln.exe"

#Loki Password Stealer
```

AT	FB	404	LPS	AM	RC	NCR	QB	RLS	NWR
✅	✅	✅	✅	✅	✅	✅	✅	✅	✅

ARTIFACTS: SYSMON EVENT ID 11 - FILE WRITE

Event 11, Sysmon

General | Details

```
File created:
RuleName: EXE
UtcTime: 2023-04-12 18:54:50.985
ProcessGuid: {16414045-fe7a-6436-7d02-000000000f00}
ProcessId: 1048
Image: C:\Users\bobby\Desktop\10343711186\ef5b66b9d3cf9a93affeeeac9b034fa87a9b30de64c4814b5da50a3db9c71e59.exe
TargetFilename: C:\Users\bobby\AppData\Local\Temp\mnsywlln.exe
CreationUtcTime: 2023-04-12 18:54:50.985
User: KILLME\bobby
```

ARTIFACTS: FILEFINDER (VELOCIRAPTOR)

```
Windows.Search.FileFinder
SearchFilesGlobTable          Glob C:\Windows\System32\config\systemprofile\**\*.dll
                              C:\Windows\System32\config\systemprofile\**\*.exe
                              C:\Windows\System32\config\systemprofile\**\*.bat
                              C:\Windows\System32\config\systemprofile\**\*.ps1
                              C:\Windows\System32\config\systemprofile\**\*.cmd
                              C:\Windows\Tasks\*.dll C:\Windows\Tasks\*.exe
                              C:\Windows\Tasks\*.bat C:\Windows\Tasks\*.ps1
                              C:\Windows\Tasks\*.cmd C:\Users\**\*.dll
                              C:\Users\**\*.exe C:\Users\**\*.bat C:\Users\**\*.ps1
                              C:\Users\**\*.cmd C:\Windows\Temp\**\*.dll
                              C:\Windows\Temp\**\*.exe C:\Windows\Temp\**\*.bat
                              C:\Windows\Temp\**\*.ps1 C:\Windows\Temp\**\*.cmd
                              C:\Windows\*.cmd C:\Windows\*.exe C:\Windows\*.bat
                              C:\Windows\*.ps1 C:\Temp\*.cmd C:\Temp\*.exe
                              C:\Temp\*.bat C:\Temp\*.ps1 C:\*\*.dll C:\*\*.exe
                              C:\*\*.bat C:\*\*.ps1 C:\*\*.cmd C:\*.exe C:\*.dll
                              C:\*.bat C:\*.ps1 C:\*.cmd
Calculate_Hash                Y
```

EXTRA MILE: C:\USERS\PUBLIC

IOC #7: DYNDNS

```
{ "Host:Port:Password":
forwarding2023.ddns.net 18114:1", [..] }
# Remcos RAT
```

AT	FB	404	LPS	AM	RC	NCR	QB	RLS	NWR
☒	☒	✅	☒	✅	✅	✅	☒	☒	✅

ARTIFACTS: DNS LOGS

PowerShell GUI

Here's how to install the DNS Server role using the Install-WindowsFeature command.

1. Run PowerShell on your computer in an elevated session.

2. To install the DNS role, run the following command. The installation doesn't require a reboot.

PowerShell Copy

```
Install-WindowsFeature -Name DNS
```

Source: learn.microsoft.com

ARTIFACTS: PASSIVE DNS

SANS Internet Storm Center

Ha

Homepage

Diaries

Podcasts

Jobs

Data

previous next

Running your Own Passive DNS Service

Published: 2019-03-27

Last Updated: 2019-03-28 06:57:53 UTC

by Xavier Mertens (Version: 1)

Source: isc.sand.edu

ARTIFACTS: DNS CACHE

```
PS C:\Users\User> Get-DnsClientCache -Type A

Entry                       RecordName                  Record Status    Section TimeTo Data   Data
                                                        Type                     Live   Length
-----                       ----------                  ------ ------    ------- ------ ------ ----
forwarding2023.ddns.net     forwarding2023.ddns.net     A      Success   Answer  603024      4 185.225.73.58
```

Windows.System.DNSCache

Type: client

Windows maintains DNS lookups for a short time in the DNS cache.

This artifact collects DNS cache entries using the WMI class MSFT_DNSClientCache.

Source: Velociraptor

IOC #8: HIGH-PORT || IP-ONLY

```
{ "C2 list":
 [ "tolatilbu.hopto.org:54984" ]}
# Netwire RAT
```

AT	FB	404	LPS	AM	RC	NCR	QB	RLS	NWR
☒	☒	☒	☒	✅	✅	✅	✅	✅	✅

ARTIFACTS: FIREWALL LOGS

MIGHT BE NOISY!

BETTER FILTER ON THE IMAGE (SYSMON EVENT ID 3) *404 Keylogger sample below*

Event 3, Sysmon

General Details

```
Network connection detected:
RuleName: Usermode
UtcTime: 2023-04-16 02:01:51.079
ProcessGuid: {f0ddfc6e-56fb-643b-bd10-000000001d00}
ProcessId: 936
Image: C:\Users\malmoeb\Desktop\bb0373bf30593baa5f750f121da34759a5c268859476fd659d241593af0c1ceb.exe
User: DESKTOP-6C00RSU\malmoeb
Protocol: tcp
Initiated: true
SourceIsIpv6: false
SourceIp: 192.168.142.2
SourceHostname: DESKTOP-6C00RSU.lan
SourcePort: 58072
SourcePortName: -
DestinationIsIpv6: false
DestinationIp: 162.159.135.233
DestinationHostname: -
DestinationPort: 443
DestinationPortName: https
```

RECAP

- Defender Exclusions
- Powershell
- Scheduled Tasks
- Run Keys
- Startup Folder
- AppData
- DNS Logs
- High Port || IP Only

CASE SOLVED: CATCHING EVIL ON THE FLY!

InfoGuard
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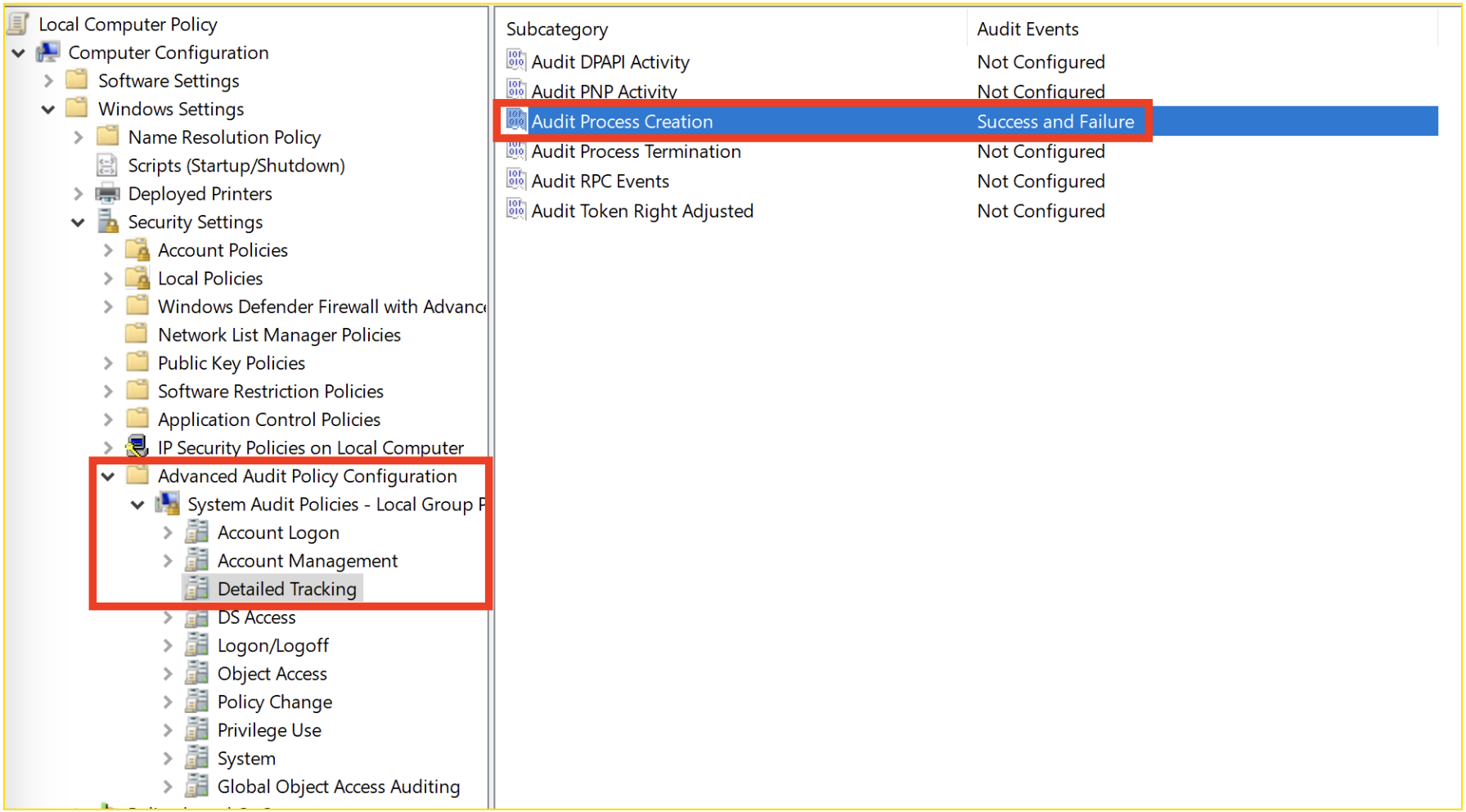
QUESTIONS?

Twitter

LinkedIn

ARTIFACTS: 4688 REGEDIT



ARTIFACTS: «ONLY» PROCESS INFORMATION

Event 4688, Microsoft Windows security auditing.

General | Details

A new process has been cr

Creator Subject:
Security ID: TOP-6C0 moeb
Account Name: b
Account Domain:
Logon ID: 0x1

Target Subject:
Security ID: D
Account Name:
Account Domain:
Logon ID: 0x0

Process Information:
New Process ID: 0x1f38
New Process Name: C:\Users\malmoeb\AppData\Local\Temp\cckgcf.exe
Token Elevation Type: %%1938
Mandatory Label: Mandatory Label\Medium Mandatory Level
Creator Process ID: 0x34c
Creator Process Name: C:\Users\malmoeb\AppData\Local\Temp\cckgcf.exe
Process Command Line:

EXTRA MILE: THREATFOX

```
curl -X POST https://threatfox-api.abuse.ch/api/v1/
-d '{ "query": "taginfo", "tag": "Netwire"}
```


```
{
    "id": "580073",
    "ioc": "51.161.104.138:5005"
    "threat_type": "botnet_cc",
    "threat_type_desc": "Indicator that identifies a botnet command&control server (C&C)",
    "ioc_type": "ip:port",
    "ioc_type_desc": "ip:port combination that is used for botnet Command&control (C&C)",
    "malware": "win.netwire",
    "malware_printable": "NetWire RC",
    "malware_alias": "NetWeird,NetWire,Recam"
    "malware_malpedia": "https:\/\/malpedia.caad.fkie.fraunhofer.de\/details\/win.netwire",
    "confidence_level": 100,
    "first_seen": "2022-05-17 08:48:34 UTC",
    "last_seen": null,
    "reference": null,
    "reporter": "abuse_ch",
    "tags": [
        "NetWire",
        "RAT"
    ]
},
```